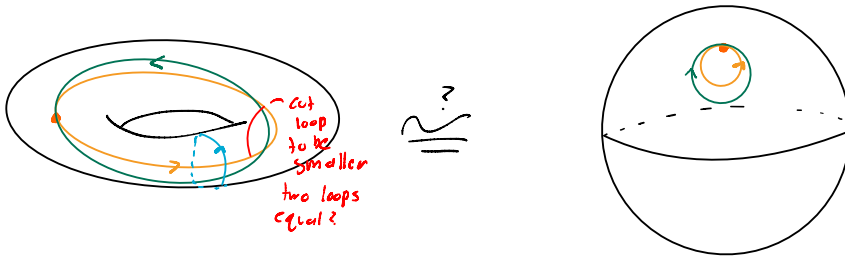
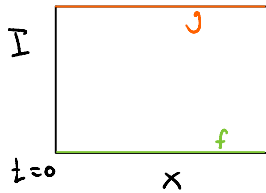


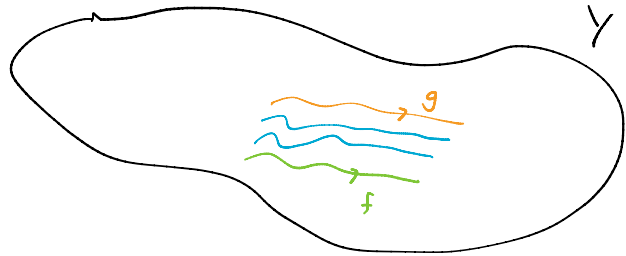
SI Homotopy Theory and the Fundamental Group



Def: IF f and g are cont maps from X to Y we say that f is homotopic to g written $f \simeq g$ if $\exists$ a cont map $F: X \times I \rightarrow Y$ st $F(x, 0) = f$ and $F(x, 1) = g(x) \forall x \in X$
 this is called a homotopy between f and g



$I = [0, 1] \subseteq \mathbb{R}$
 subspace top

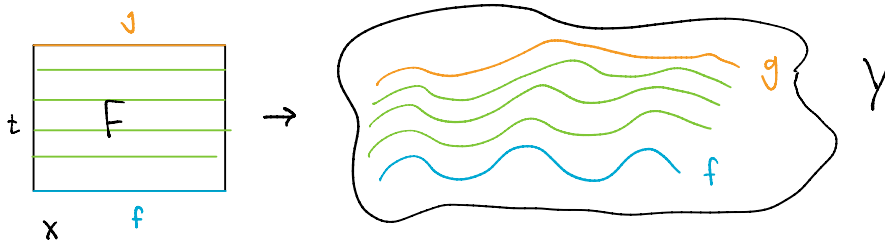


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Homotopy

$I = [0, 1] \subseteq \mathbb{R}$

Def: $f, g: X \rightarrow Y$ cont we say f is homotopic to g written $f \simeq g$ if $\exists$ cont map $F: X \times I \rightarrow Y$ st $F(x, 0) = f(x)$ $F(x, 1) = g(x) \forall x \in X$

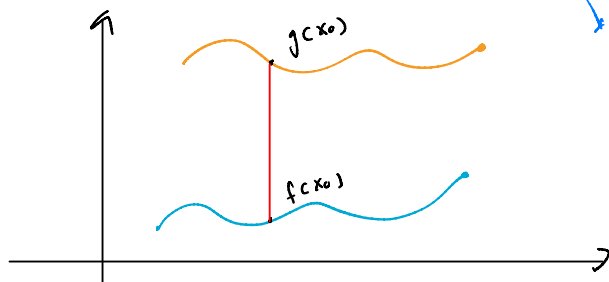


We say f is nullhomotopic if f is homotopic to a constant map

Ex ① Let $f, g: X \rightarrow \mathbb{R}^n$ cont

Then $f \simeq g$ A homotopy can be defined by

$$F(x, t) = (1-t)f(x) + t(g(x))$$



Straight line homotopy

$\therefore$ Any map $f: X \rightarrow \mathbb{R}^n$ is nullhomotopic

Ex ② $i: S^1 \rightarrow \mathbb{R}^2 - \{0, 0\}$
 Circle punctured plane

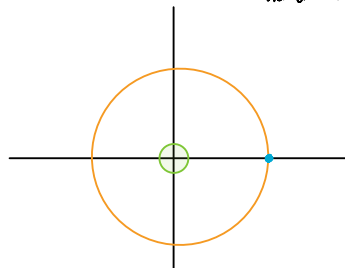
$i(x) = x$ inclusion map

$f: S^1 \rightarrow \mathbb{R}^2 - \{0, 0\}$

$f(x) = (1, 0)$

Then $i \neq f$

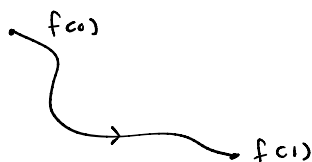
Seems clear
hard to show



A path is a cont map $f: I \rightarrow X$. We call $f(0)$ the initial point and $f(1)$ the terminal pt of f



$\xrightarrow{f}$



Def: Two paths $f, g: I \rightarrow X$ are path homotopic written $f \simeq_p g$ if

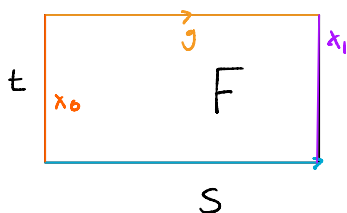
① $f(0) = g(0) = x_0$

② $f(1) = g(1) = x_1$

③ $\exists$ (a homotopy) cont map $F: I \times I \rightarrow X$ such that

$$F(s, 0) = f(s) \quad F(s, 1) = g(s) \quad \forall s \in I$$

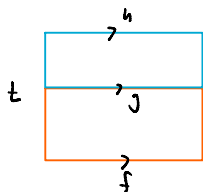
$$F(0, t) = x_0 \quad F(1, t) = x_1 \quad \forall t \in I$$



Lemma: The relations $\simeq$ and $\simeq_p$ are equivalence relations.

Proof: $\cdot f \simeq f \quad F(x, t) = f(x)$

$\cdot$ Suppose $g \simeq f$ Let $F: X \times I \rightarrow Y$ be a homotopy from f to g then $G: X \times I \rightarrow Y$ defined by $G(x, t) = F(x, 1-t)$ is a homotopy from g to f . Suppose $f \stackrel{F}{\simeq} g \quad g \stackrel{G}{\simeq} h$ show $f \simeq h$

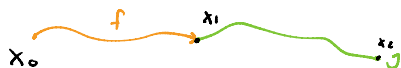


$$H(x, t) = \begin{cases} F(x, 2t) & 0 \leq t \leq 1/2 \\ G(x, t-1/2) & 1/2 \leq t \leq 1 \end{cases}$$

Homotopy from f to h (pasting lemma)

If $f: I \rightarrow X$ is a path in X let $[f]$ denote the path equivalence class of f

Product of Paths



If f is a path in X from x_0 to x_1 and g is a path from x_1 to x_2

then define $f * g: I \rightarrow X$ by $(f * g)(s) = \begin{cases} f(2s) & 0 \leq s \leq 1/2 \\ g(2s-1) & 1/2 \leq s \leq 1 \end{cases}$

path from x_0 to x_2

concatination of paths

Thm: The operation $*$ is well defined on path homotopy classes
The following hold

① Associativity $[f] * ([g] * [h]) = ([f] * [g]) * [h]$

Define $[f] * [g] := [f * g]$

② (Right and Left Identities)

Given $x \in X$ let e_x denote the constant map $e_x(s) = x \quad \forall s \in I$

If f is a path in X from x_0 to x_1 , then $[f] * [e_{x_1}] = [f]$

$[e_{x_0}] * [f] = [f]$

③ (Inverses) Given a path $f: I \rightarrow X$,

define $\bar{f}: I \rightarrow X$ by $\bar{f}(s) = f(1-s)$ go opposite direction

Then $[f] * [\bar{f}] = [e_{x_0}] \quad [\bar{f}] * [f] = [e_{x_1}]$